

Volatility Forecasting: A Statistical and Deep Learning Approach

Abstract

1 Introduction

1.1 Context

Stock markets originated at 1602, in Amsterdam, Dutch Republic (now the Netherlands). In that year, the Dutch East India Company issues transferable shares, and investors began trading them with one another. At that point in time, it originated the possibility to earn money by buying the paper at a lower price, hoping that the company would grow with its expansion. Or, if someone had privileged information, e.g. a failed expedition or a successful one, would be in a better position than the others, being able to sell its position before it loses value or buy more before it gets a higher price.

The desire to obtain and interpret information that may affect stock prices remains central to financial markets. Services such as the Bloomberg Terminal give investors rapid access to news and data, while some traders use direct exchange feeds when even small delays matter. If you receive the information before every player, you have a leverage over them and can position yourself in a beneficial way. Jim Simons, the founder of Renaissance Technologies, whose fund Medallion had an average annual return of 66% before investor fees, started his trading using news in early human trading decisions, before passing to models who studies historical market data as stated by Gregory Zuckerman in his Jim Simons biography, "The Man Who Solved the Market".

As competition for information increases, the opportunity to profit from that same information decreases. This process leads to an efficient market. The efficient market theory was developed by Eugene Fama, which states: "Prices reflect available information". An efficient market is one where prices

adjust quickly enough to available information such that investor cannot reliably earn extra returns from it after accounting for risk and trading costs. As markets converges to an extremely high degree of efficiency, one can assume that analyzing the prices historical data provides the answer to profit. As described by Gregory Zuckerman, Simons was searching historical price data for a recurring pattern.

A historical data is known as a time series. Time series analysis has been a widely explored field in statistics and ML (Machine Learning). Statistical analysis is focused on understanding the properties of the data (stationarity, mean reversion, seasonality, trend, cycles, etc.) and creating a model, based on such characteristics that sucessfully explains the movements observed in such data. ML, in the meanwhile, is interested in creating a model that efficiently explains the data without effectively needing its traits.

An early landmark of time series statistical modelling is with Yule's work in 1927 where he introduced the AR (Autoregressive) model, where current value depends on its past values.

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